

PV–PORT 180 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P100/110, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

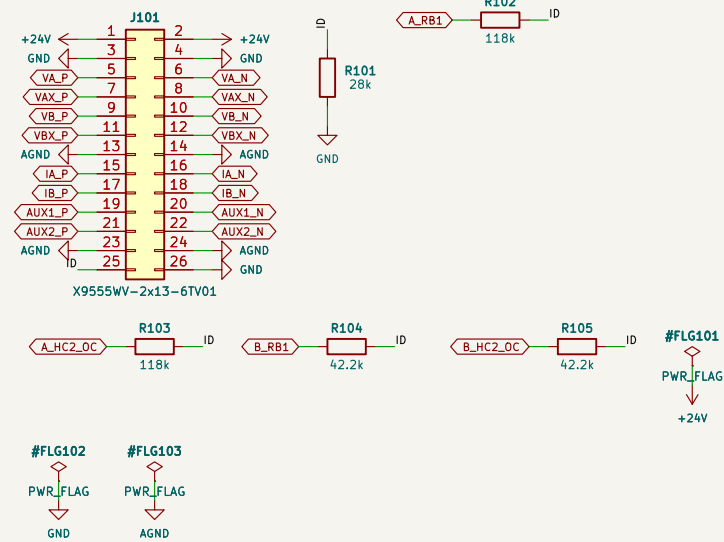
Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

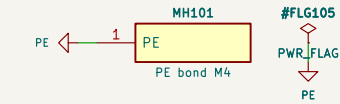
SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



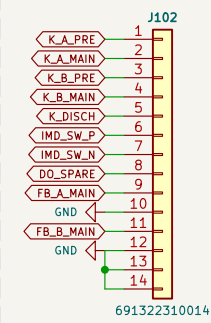
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV-PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): 195 states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4); CTRL decodes them with its port-only table.

PE BOND



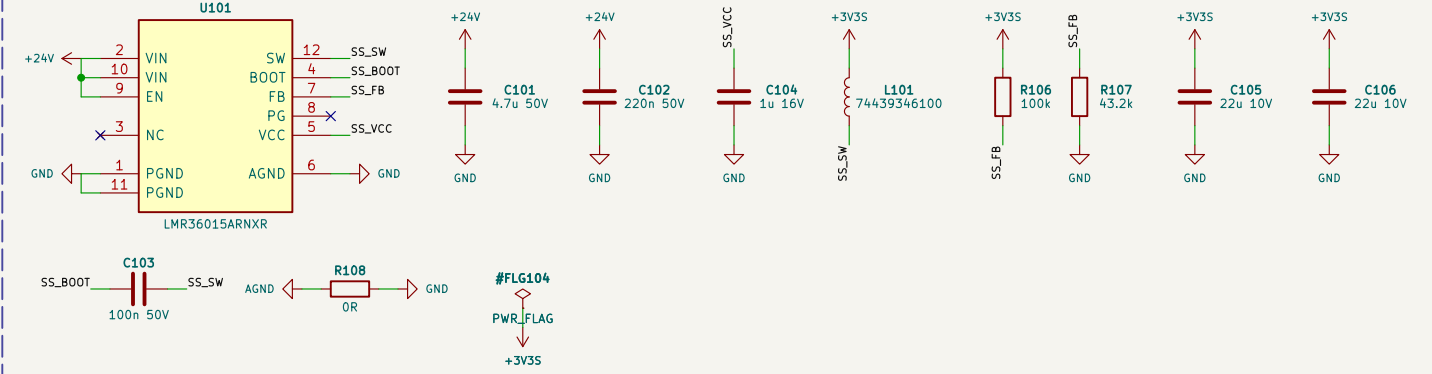
Board PE to chassis at the SPD / Y-capacitor end.

CONTROL CONNECTOR (SYS-IO-AUX)



port.CTRL_PINS: DO1-DO8 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS DO4 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



LMR36015 400 kHz, 24 V -> 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), <= 0.34 A for 6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
 Sheet 2
 Schematics authored by Chinmoy Bhuyan
 HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /01 Connectors and sensor supply/
File: 01_io.kicad_sch

Title: PV-PORT 180 A port board – Connectors and sensor supply

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/7

[illegible][illegible]

The schematic diagram illustrates the power distribution network (PDN) for the REF3030E and REF3030E+0.1% network. It shows the power and ground connections for the REF3030E and REF3030E+0.1% network, including the power supply, decoupling capacitors, and the network components. The REF3030E is shown in a yellow box, and the REF3030E+0.1% network is shown in a green box. The diagram includes labels for various components, such as capacitors (C328, C329, C330, C331), resistors (R318, R320, R321, R322, R324, R326, R327, R328, R330), and the network components (A_HCL_PA, A_HCL_PB, A_HCL_PC, A_HCL_PD, A_HCL_PE, A_HCL_PF, A_HCL_PG, A_HCL_PH, A_HCL_PI, A_HCL_PJ, A_HCL_PK, A_HCL_PL, A_HCL_PM, A_HCL_PN, A_HCL_PO, A_HCL_PP, A_HCL_PQ, A_HCL_PR, A_HCL_PS, A_HCL_PT, A_HCL_PU, A_HCL_PV, A_HCL_PW, A_HCL_PX, A_HCL_PY, A_HCL_PZ). The diagram also includes a table of component values and a table of component types.

Component	Value	Type
C328	100n 50V	Capacitor
C329	100n 50V	Capacitor
C330	100n 50V	Capacitor
C331	100n 50V	Capacitor
R318	49.9k	Resistor
R320	86.6k	Resistor
R321	49.9k	Resistor
R322	88.7k	Resistor
R324	49.9k	Resistor
R326	86.6k	Resistor
R327	49.9k	Resistor
R328	88.7k	Resistor
R330	100k	Resistor

Component	Value	Type
A_HCL_PA	49.9k	Resistor
A_HCL_PB	49.9k	Resistor
A_HCL_PC	49.9k	Resistor
A_HCL_PD	49.9k	Resistor
A_HCL_PE	49.9k	Resistor
A_HCL_PF	49.9k	Resistor
A_HCL_PG	49.9k	Resistor
A_HCL_PH	49.9k	Resistor
A_HCL_PI	49.9k	Resistor
A_HCL_PJ	49.9k	Resistor
A_HCL_PK	49.9k	Resistor
A_HCL_PL	49.9k	Resistor
A_HCL_PM	49.9k	Resistor
A_HCL_PN	49.9k	Resistor
A_HCL_PO	49.9k	Resistor
A_HCL_PP	49.9k	Resistor
A_HCL_PQ	49.9k	Resistor
A_HCL_PR	49.9k	Resistor
A_HCL_PS	49.9k	Resistor
A_HCL_PT	49.9k	Resistor
A_HCL_PU	49.9k	Resistor
A_HCL_PV	49.9k	Resistor
A_HCL_PW	49.9k	Resistor
A_HCL_PX	49.9k	Resistor
A_HCL_PY	49.9k	Resistor
A_HCL_PZ	49.9k	Resistor

||A| > 410 A (band 382-44k A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND
 ROAD = valie -> HOLD1, Test Injection 86.6 k from A_HCL_TST (side A) / A_HCL_TST (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

$|I_A| > 410 \text{ A (band 382-441 A, own REF3030E} \pm 0.1 \% \text{ network CM-cancelling) AND FB = closed AND}$
 $(I_{OAC} = \text{valid} \rightarrow \text{HOLD2: Test Injection } 86.6 \text{ A from A_TST_T2A (side A) / A_TST_T2B (side B): shifts the threshold}$
 $\text{by } 459 \text{ A, so a healthy side trips at } 0 \text{ A under TEST only.}$

100 k / 15 k → Schmitt buffer T (ch1 side A), T → 47 k / 22 uF → T_d (ch1 side B). One-shots (3.3 uF / 47 k) of T → T2A (ch2 side A) and of T_d → T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4c windows, what it finds and what it does not).

[illegible]

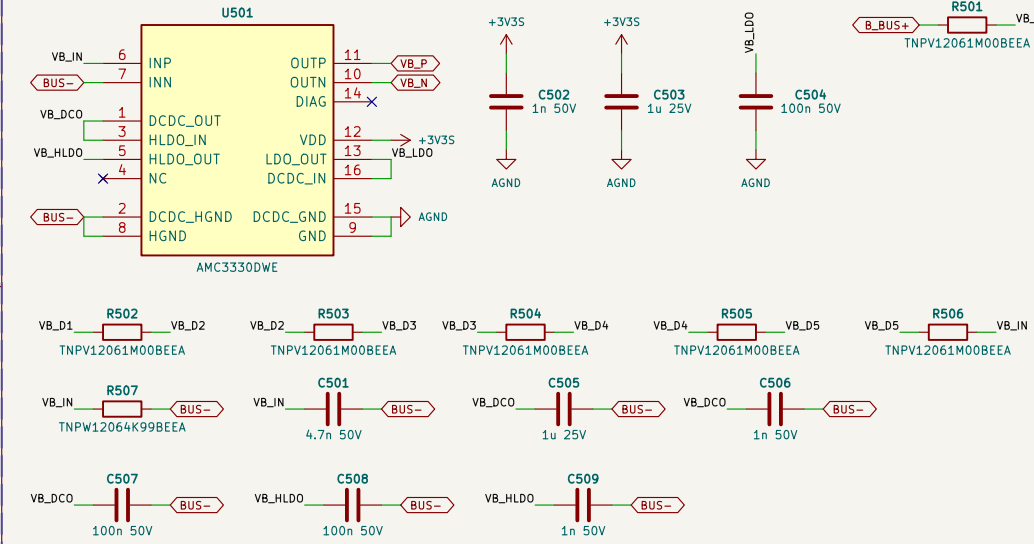
Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 4
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /03 Port A sensing and hold logic/
File: 03_portA_sense.kicad_sch

Title: *PV-PORT 180 A port board – Port A sensing and hold logic*

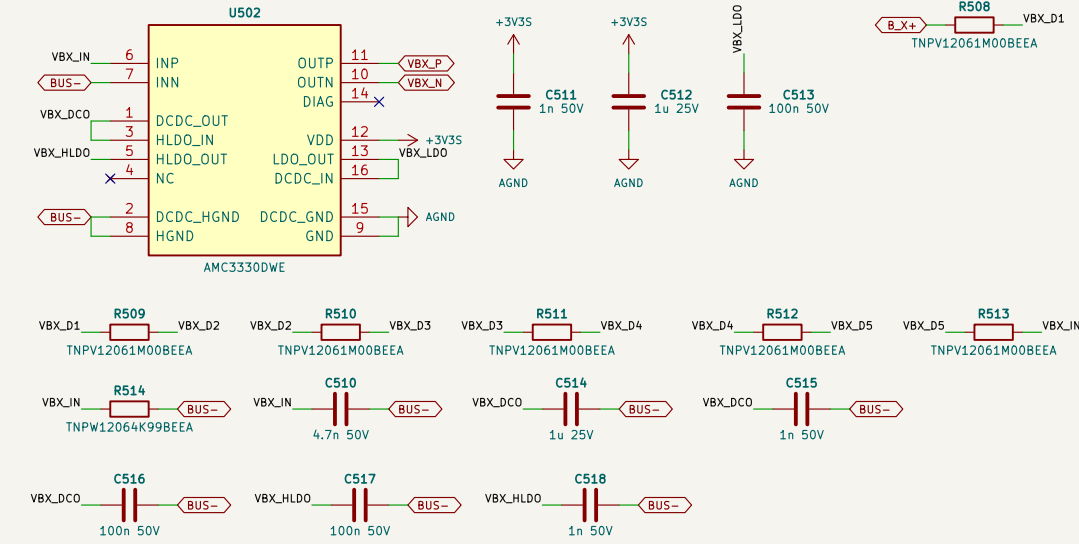
Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 4/7

VB – BUS-SIDE VOLTAGE



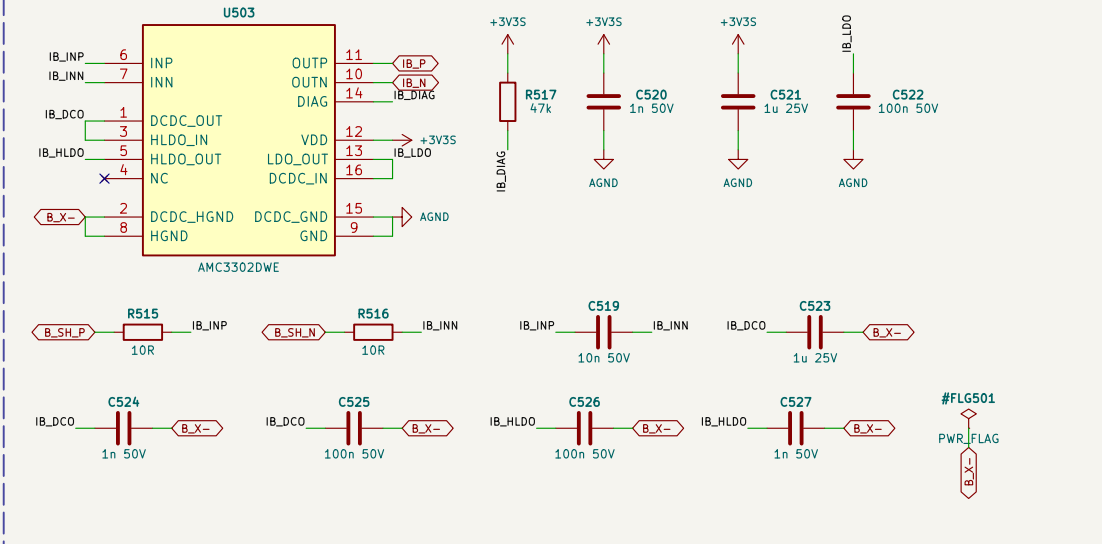
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V. 4.7 nF: lag <= 27 us incl. AMC3330. Out: VB_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

VBX – TERMINAL-SIDE VOLTAGE



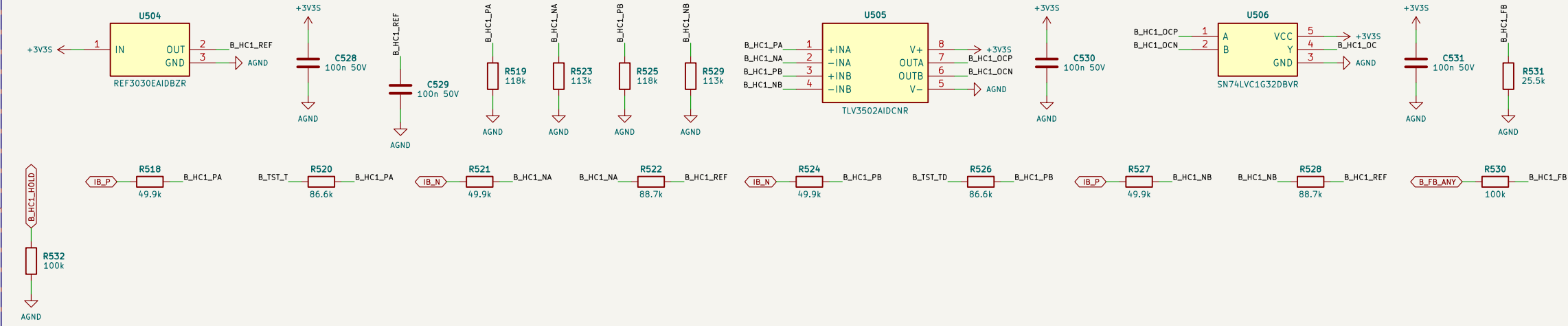
6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
 <= 27 us incl. AMC3330. Out: VBX_P/N (+/-2 V on 1.44 V, PORT contract). HGND = BUS-.

IB – PORT CURRENT



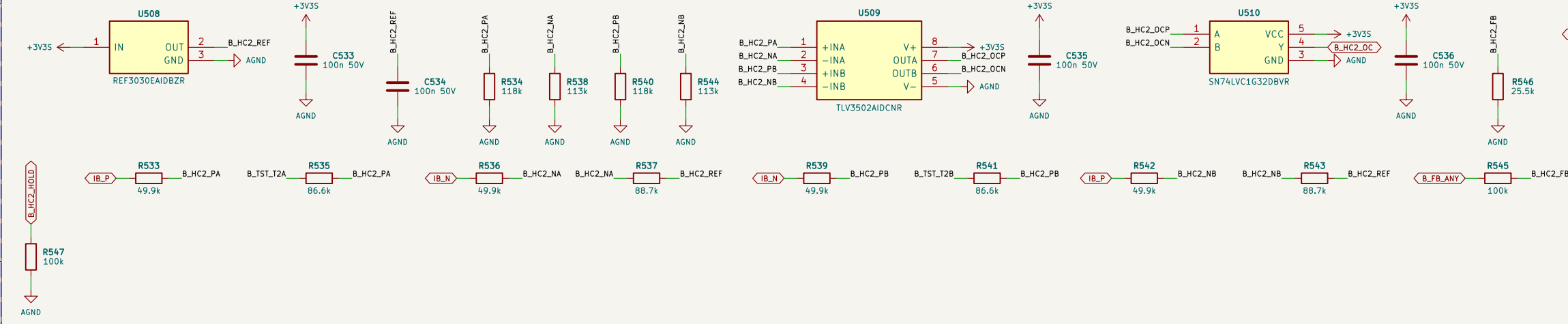
Linear ± 500 A, clip ± 640 A (OC trip 270 A, SC trip 450 A inside). 10 ohm + 10 nF input filter (SBASA11B Fig 8-2). HGND on the shunt's X- bolt by its own trace. Fail-safe out = -2.57 V.

HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



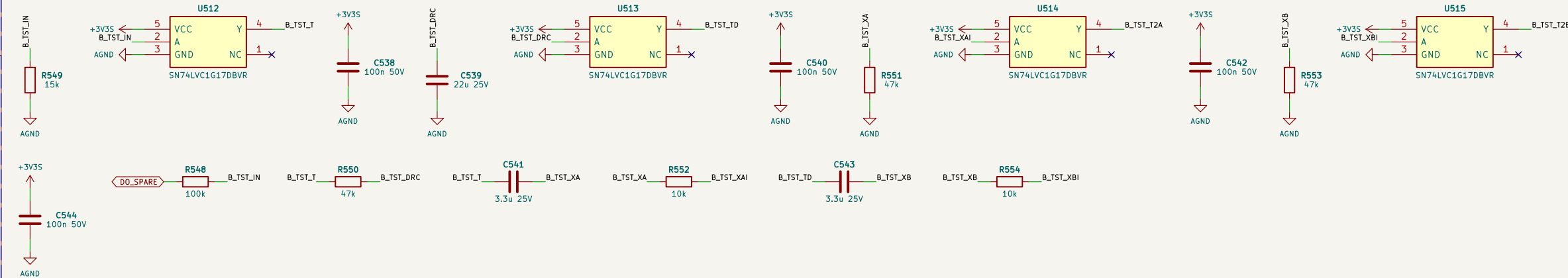
|IB| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD1. Test injection 86.6 k from B_TST_T (side A) / B_TST_TD (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



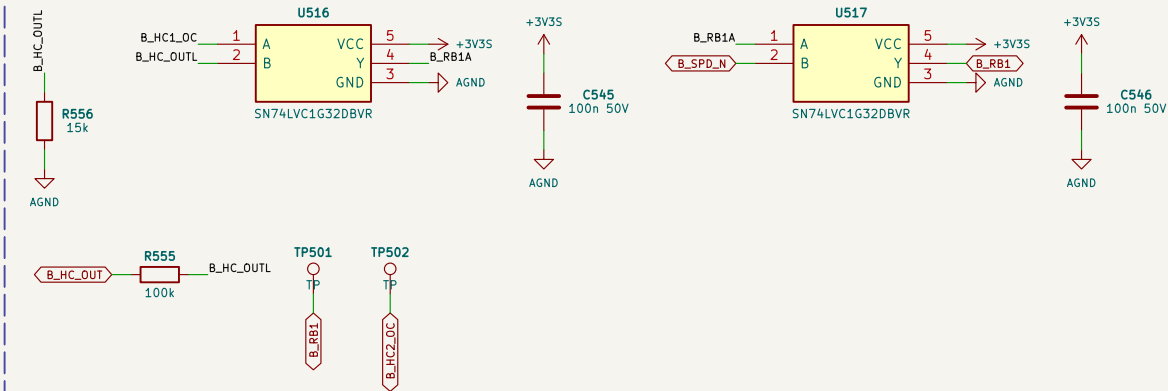
|IB| > 410 A (band 382-441 A, own REF3030E + 0.1 % network, CM-cancelling) AND FB = closed AND DIAG = valid -> HOLD2. Test injection 86.6 k from B_TST_T2A (side A) / B_TST_T2B (side B): shifts the threshold by 459 A, so a healthy side trips at 0 A under TEST only.

PROOF-TEST INJECTION (DO_SPARE = TEST)



100 k / 15 k $\rightarrow$ Schmitt buffer T (ch1 side A). T $\rightarrow$ 47 k / 22 μ F $\rightarrow$ T_d (ch1 side B). One-shots (3.3 μ / 47 k) of T $\rightarrow$ T2A (ch2 side A) and of T_d $\rightarrow$ T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4: windows, what it finds and what it does not).

READBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



RB1 = OC1 OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
RB2 = OC2. PV-PORT sums them onto the ID line (board_io); firmware reads RB1 + RB2 against the measured current in operation and through the proof-test windows at start.

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 6
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

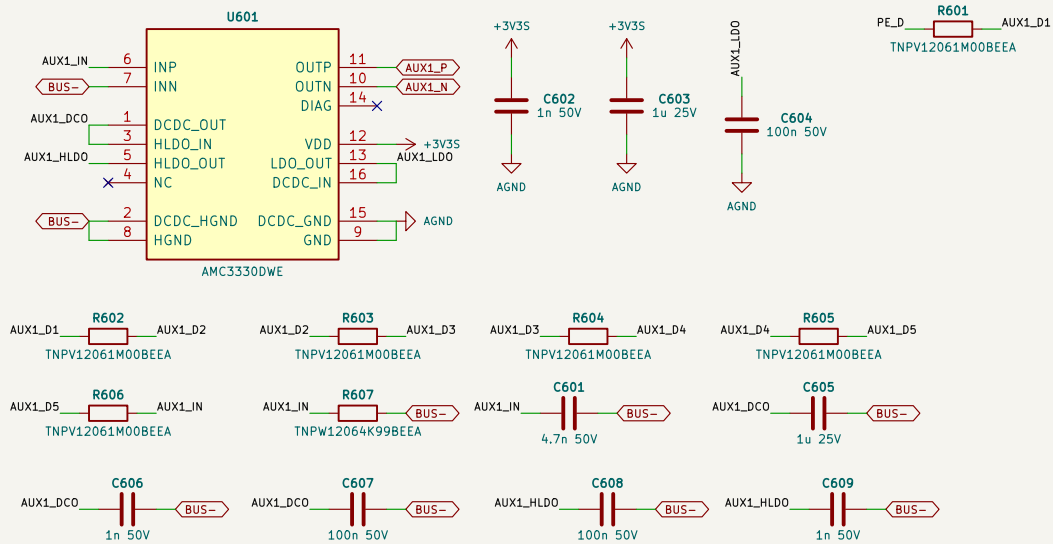
Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

Title: PV-PORT 180 A port board – Port B sensing and hold logic

Size: A2	Date: 2026-10-04	Rev: F0
KiCad EDA 10.0.6		Id: 6/7

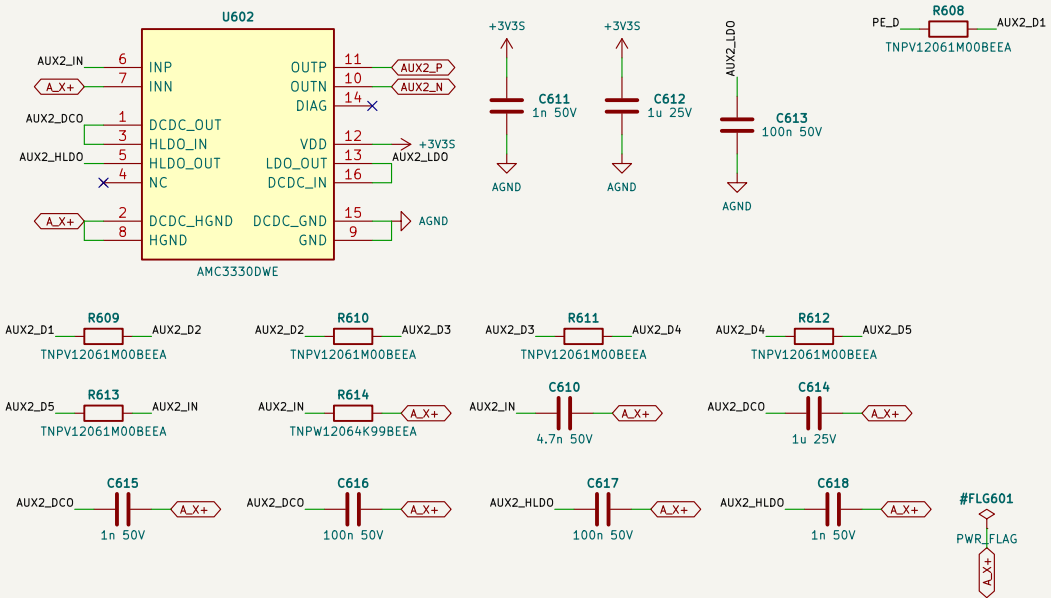
Rev: F0
Id: 6 / 7

AUX1 – PE RELATIVE TO BUS–



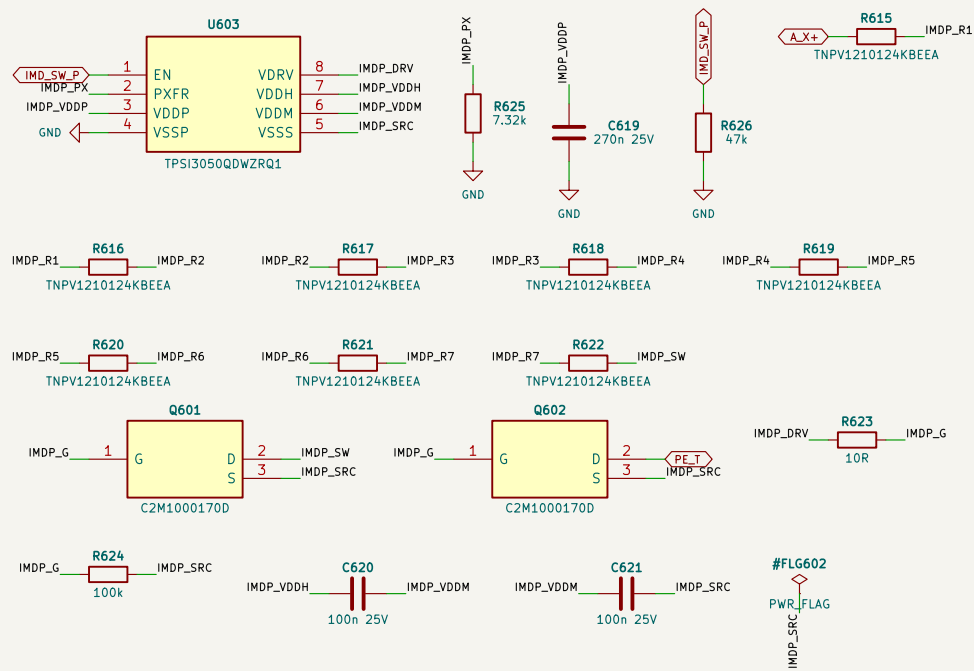
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

AUX2 – PE RELATIVE TO A_X+



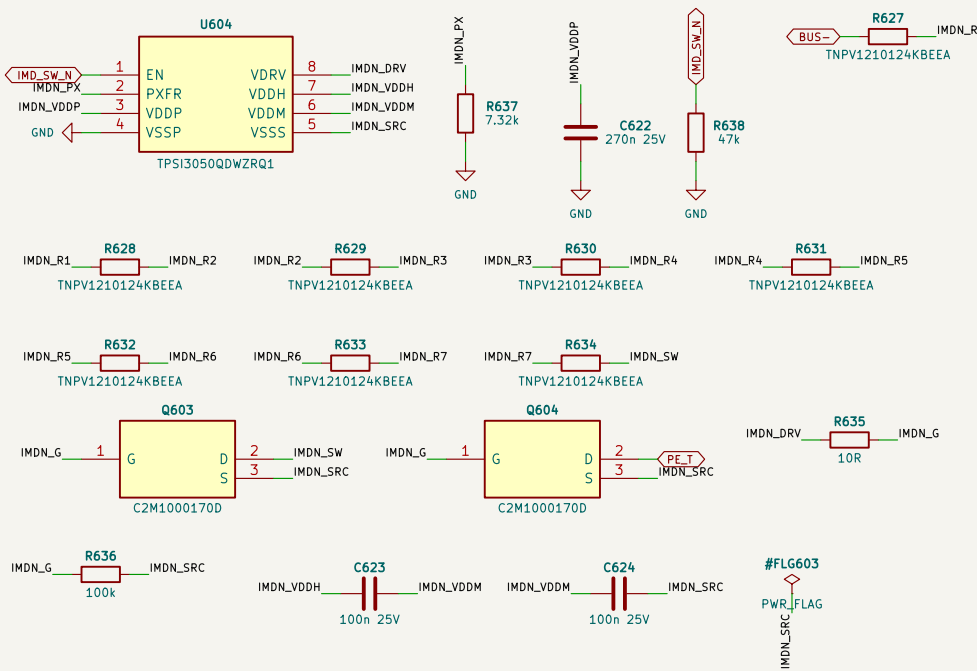
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



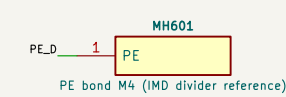
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

TEST STRING BUS– -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.